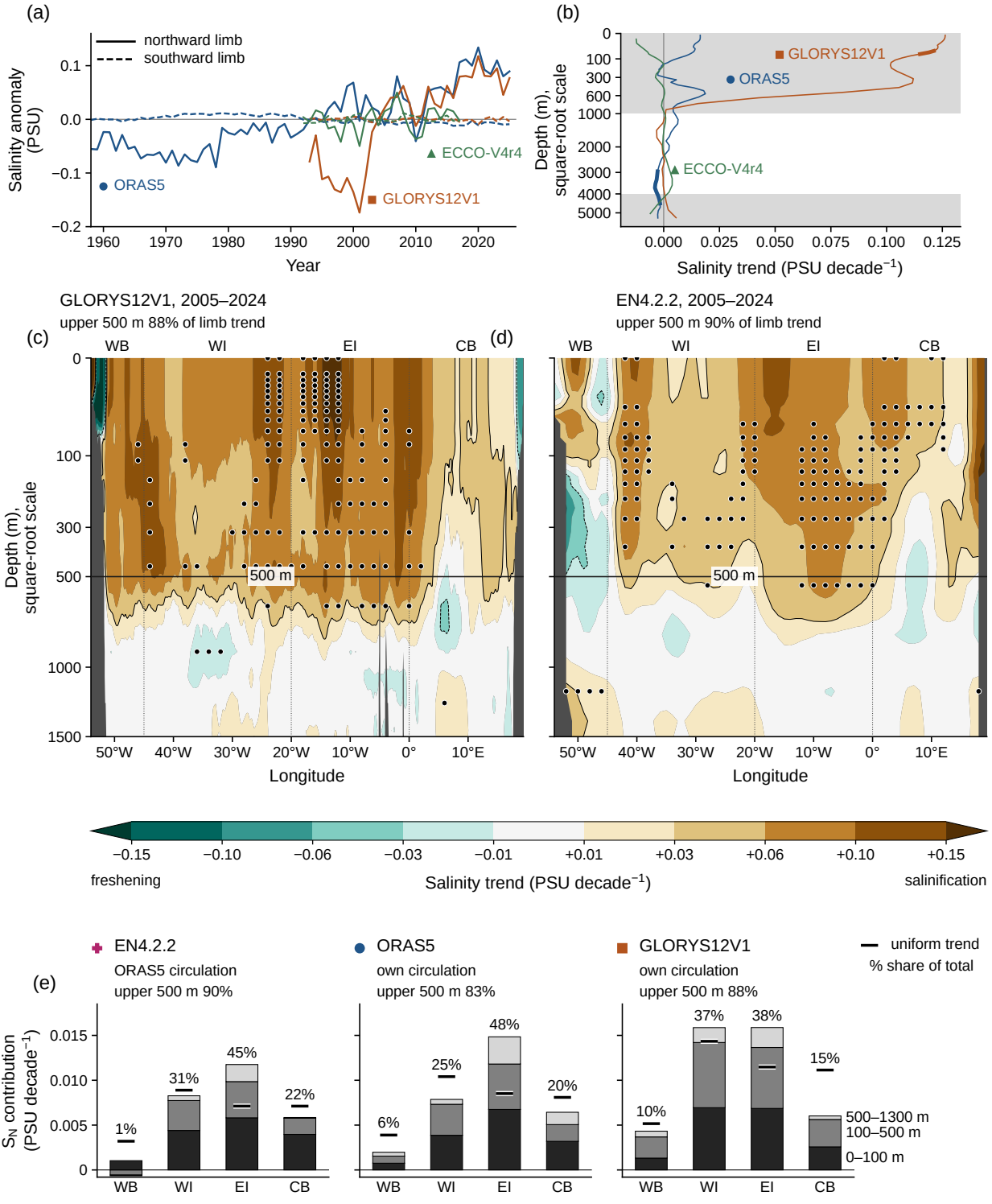

(a)
northward limb
southward limb
Salinity anomaly (PSU)
ORAS5
ECCO-V4r4
GLORYS12V1
Year
(b)
GLORYS12V1
ORAS5
ECCO-V4r4
Depth (m), square-root scale
Salinity trend (PSU decade−1)
GLORYS12V1, 2005–2024
upper 500 m 88% of limb trend
EN4.2.2, 2005–2024
upper 500 m 90% of limb trend
(c)
(d)
WB
WI
EI
CB
500 m
Longitude
−0.15
−0.10
−0.06
−0.03
−0.01
+0.01
+0.03
+0.06
+0.10
+0.15
freshening
salinification
(e)
EN4.2.2
ORAS5 circulation
upper 500 m 90%
ORAS5
own circulation
upper 500 m 83%
GLORYS12V1
own circulation
upper 500 m 88%
uniform trend
% share of total
SN contribution (PSU decade−1)
1%
31%
45%
22%
6%
25%
48%
20%
10%
37%
38%
15%
500–1300 m
100–500 m
0–100 m